

```

29     const response = await fetch(
30         'https://nominatim.openstreetmap.org/reverse' +
31         '?format=json&lat=${lat}&lon=${lng}'
32     );
33     const data = await response.json();
34     return data.display_name || `${lat}, ${lng}`;
35 } catch {
36     return `${lat.toFixed(6)}, ${lng.toFixed(6)}`;
37 }
38 }
39
40 export function calculateDistance(
41     loc1: Location, loc2: Location
42 ): number {
43     const R = 6371; // Earth radius in km
44     const dLat = ((loc2.latitude - loc1.latitude) * Math.PI) / 180;
45     const dLon = ((loc2.longitude - loc1.longitude) * Math.PI) /
46         180;
47     const a = Math.sin(dLat / 2) * Math.sin(dLat / 2) +
48         Math.cos((loc1.latitude * Math.PI) / 180) *
49         Math.cos((loc2.latitude * Math.PI) / 180) *
50         Math.sin(dLon / 2) * Math.sin(dLon / 2);
51     const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
52     return R * c;
53 }

```

Listing 8.7: Location Service (location-service.ts)

The location service utilizes the browser's Geolocation API for acquiring GPS coordinates and OpenStreetMap's Nominatim service for reverse geocoding. The `calculateDistance` function implements the Haversine formula to compute the great-circle distance between two geographic points, which is used to sort nearby waste listings for collection agents.

8.8 Notification Service

The notification service manages in-app notifications for status updates and system events:

```

1 import { collection, addDoc, query, where, orderBy,
2     getCountFromServer, updateDoc, doc,
3     serverTimestamp } from 'firebase/firestore';
4 import { db } from '../firebase';
5
6 export interface AppNotification {
7     id?: string;

```